

MID-OHIO VALLEY, WV, USA

WMO: 724273

Lat: 39.345N

Lon: 81.439W

Elev: 253

StdP: 98.32

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 03804

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-14.0	-11.1	-19.5	0.7	-12.2	-16.8	0.9	-9.1	9.3	3.0	8.5	2.3	2.2	260	0.437

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.9	32.5	23.2	31.2	22.6	30.0	22.0	24.8	30.0	24.0	28.8	23.4	27.9	3.1	260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
23.2	18.6	27.2	22.5	17.8	26.5	21.9	17.1	25.8	76.4	30.2	73.2	28.9	70.5	27.7	28.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
8.2	7.2	6.1	-18.5	34.3	4.0	1.8	-21.3	35.5	-23.6	36.6	-25.8	37.6	-28.7	38.9		
			-18.9	26.4	3.8	1.0	-21.6	27.1	-23.9	27.7	-26.0	28.3	-28.8	29.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	12.4	0.0	1.8	6.3	12.8	17.4	21.6	23.5	22.9	19.5	13.0	7.0	2.5
	DBStd	9.65	7.01	6.29	6.02	5.23	4.37	3.12	2.60	2.65	3.77	4.83	5.26	5.69
	HDD10.0	1124	321	237	148	31	2	0	0	0	0	24	121	241
	HDD18.3	2706	569	462	375	178	72	9	1	2	31	176	341	491
	CDD10.0	2005	9	8	33	113	233	348	418	401	284	117	30	9
	CDD18.3	544	0	0	2	11	43	107	161	144	65	11	1	0
	CDH23.3	4673	0	1	18	131	417	912	1332	1183	574	103	4	0
	CDH26.7	1497	0	0	2	22	107	287	468	402	188	21	0	0
Wind	WSAvg	2.5	3.1	3.1	3.1	3.1	2.3	2.1	1.8	1.7	1.8	2.2	2.6	2.8
Precipitation	PrecAvg	1055	66	64	103	87	101	97	108	106	81	74	85	81
	PrecMax	1425	150	136	238	193	266	187	203	211	167	200	200	206
	PrecMin	692	17	14	25	21	22	11	32	12	15	12	12	11
	PrecStd	176	32	33	52	42	56	45	47	47	37	45	39	43
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	19.7	21.1	25.9	29.1	31.0	33.0	34.0	34.0	33.4	29.0	23.4	19.4
		MCWB	14.6	13.7	15.7	17.6	21.0	22.9	23.7	23.7	21.7	20.5	16.3	14.8
	2%	DB	16.4	17.4	22.5	26.8	29.3	31.2	32.3	32.2	31.0	26.1	20.8	16.6
		MCWB	12.7	11.7	13.9	16.1	20.3	22.7	23.9	23.2	21.2	18.5	14.3	12.5
	5%	DB	13.6	14.6	19.4	24.6	27.8	29.8	30.9	30.6	28.8	23.8	18.2	14.0
		MCWB	11.0	10.3	11.9	15.3	19.7	22.1	23.1	22.6	20.6	17.0	12.8	10.6
	10%	DB	10.6	11.8	16.4	22.3	26.0	28.3	29.3	29.0	26.8	21.5	16.0	11.6
		MCWB	7.5	8.1	11.1	14.1	18.9	21.4	22.4	21.9	19.7	16.1	11.6	8.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	15.4	15.4	16.8	19.5	23.0	25.0	26.1	25.7	23.8	21.7	17.7	15.4
		MCDB	18.4	19.2	23.4	26.6	28.7	30.5	31.6	30.7	29.0	26.9	21.7	18.4
	2%	WB	13.3	13.1	15.1	17.9	21.8	23.9	25.0	24.7	22.7	20.1	15.4	13.5
		MCDB	15.6	15.7	20.4	23.7	27.3	29.1	30.3	29.7	28.0	24.3	19.1	15.8
	5%	WB	11.3	10.9	13.7	16.6	20.7	23.1	24.2	23.8	21.8	18.4	13.8	11.3
		MCDB	13.7	14.0	17.8	22.0	25.8	28.1	28.7	28.5	26.5	22.1	17.1	13.4
	10%	WB	7.8	8.1	11.7	15.4	19.8	22.3	23.5	23.0	20.8	16.9	12.0	8.8
		MCDB	10.0	11.3	15.9	20.5	24.4	27.0	27.7	27.2	24.7	20.6	15.7	11.3
Mean Daily Temperature Range	5% DB	MDBR	8.4	9.5	11.0	12.8	11.9	11.2	10.9	11.3	12.0	11.8	10.4	8.3
		MCDBR	10.5	13.9	15.5	15.8	14.3	13.0	12.4	12.9	14.6	14.6	13.9	11.9
		MCWBR	7.9	10.1	8.8	8.0	6.7	5.9	5.2	5.2	6.0	7.6	8.8	8.9
	5% WB	MCDBR	9.7	12.4	12.6	12.6	11.7	11.1	10.4	10.7	11.7	12.1	11.2	10.4
		MCWBR	7.9	10.1	8.2	7.3	5.9	5.5	4.9	5.0	5.4	6.8	8.3	8.8
Clear-Sky Solar Irradiance	taub	0.307	0.321	0.346	0.389	0.434	0.462	0.483	0.484	0.407	0.364	0.324	0.299	
	taud	2.473	2.421	2.414	2.314	2.198	2.160	2.119	2.092	2.297	2.419	2.506	2.532	
	Ebn at Noon	867	901	911	887	850	823	801	788	834	839	839	852	
	Edh at Noon	79	95	106	125	144	150	155	155	119	95	76	70	
All-Sky Solar Radiation	RadAvg	1.69	2.43	3.53	4.71	5.42	5.97	5.71	5.35	4.42	3.08	2.14	1.45	
	RadStd	0.18	0.35	0.29	0.43	0.45	0.42	0.35	0.32	0.43	0.34	0.28	0.18	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (32 neighbors)	+0.44	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+135	+69

Nomenclature: See separate page